

Work experience (related)

08/2024–07/2025 | **Bioinformatician**

Erasmus University Medical Center, Rotterdam, NL

- Engineered scalable ETL pipelines integrating heterogeneous, multi-modal datasets (multi-omics) and high-throughput inputs.
- Architected High-Performance Computing (HPC) workflows to process terabytes of unstructured sequence data, optimizing resource allocation using Slurm.

08/2020–07/2024 | **Bioinformatician**

Amsterdam University Medical Center, Cancer Center Amsterdam, Department of Pathology, Amsterdam, NL

- Developed and optimized predictive Machine Learning models (classification and regression) to detect anomalies in noisy signal data, achieving state-of-the-art accuracy.
- Led the engineering and analysis of complex sequence patterns, translating raw unstructured data into actionable structured features.

09/2019–08/2021 | **Bioinformatics consultant**

Griffsoft Zrt., Szeged, HU

- Advised a software engineering team, bridging the gap between domain requirements (biotechnology - software development) and algorithm implementation for commercial analytics software.

Affiliations

11/2023–08/2024 | **Bioinformatician**

University of Manchester, Manchester, UK

08/2024–12/2024 | **Bioinformatician**

Amsterdam University Medical Center, Amsterdam, NL

08/2025–12/2025 | **Bioinformatician**

Erasmus University Medical Center, Rotterdam, NL

Work experience (other)

2016 | **Research assistant**

University of Szeged, Department of Medical Biology, Szeged, HU

2013–2016 | **Encounter group trainer**

Nagyító Egyesület, HU

2013–2015 | **Secondary education teacher**

Lakatos Menyhért Általános Iskola és Gimnázium, Budapest, HU

2011–2013 | **Secondary education teacher**

Jósika Miklós Elméleti Líceum, Turda, RO

2008–2013 | **Encounter group trainer**

Kolozsvári Nagyító Egyesület (Alteredu), Cluj-Napoca, RO

Education and training

2016–2020 | **PhD in Medical Genetics**

University of Szeged, Faculty of General Medicine, Szeged, HU

M.Sc. in Biology (2010-2012) | **B.Sc. in Biology** (2007-2010)

Babes-Bolyai University, Cluj-Napoca, RO

**Norbert Moldován****SENIOR
DATA SCIENTIST**

Highlights

10+ years of research experience on high-dimensional, unstructured data.

Expert in Deep Learning, Predictive Modeling, and building scalable HPC workflows.

Proficient in Python and R for production-grade ML pipelines (PyTorch, Scikit-learn, Snakemake).

Traditional and vibe coder.

22 publications in Q1 journals
| 4 grants | 2 awards |
International speaker.

Personal details

Date of birth: 06/02/1988

Nationality: Hu, Ro

Residence: Amsterdam, NL

I do not require a Visa.

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Mother tongue: Hungarian

Foreign languages	Understanding		Speaking		Writing
	Listening	Reading	Spoken interaction	Spoken production	
English	C2	C2	C2	C2	C2
Romanian	C1	C1	C1	C1	C1
Dutch	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user
Common European Framework of Reference for Languages

Skills and competences

High-Volume Data Processing

- Expertise in processing raw unstructured sequence data.
- Proficient in time-series analysis and signal Processing.

Machine Learning & AI

- Knowledgeable with Scikit-learn, PyTorch and TensorFlow.
- Natural Language Processing (NLP) techniques (transformers, diffusion models).

Scalability and reproducibility

- Proficient in Snakemake, Anaconda/Mamba, Docker, Git for scalable and reproducible pipelines.
- Knowledgeable with ML hyper-parameter optimization with Optuna.

High-Performance Computing & Parallelization

- Experienced in optimizing bioinformatics pipelines for multi-threading and GPU acceleration.
- Familiar with Dask for large-scale data analysis.

Communication and management

- Experienced in delivering complex concepts.
- Skilled in active listening to facilitate discussions and enhance learning outcomes.
- Confident in assertive communication, ensuring clarity and effectiveness in professional interactions.
- Proven organizational skills through planning and facilitating encounter and self-knowledge groups.
- Experienced in team leadership, guiding diverse groups as a trainer and educator.

Data management and FAIR principles

- Experienced with implementing the FAIR principles (Findable, Accessible, Interoperable, Reusable).



Research & Innovation Highlights

1. [PMID: 38128532](#) | Architected "FrEIA," a scalable Python-based pipeline for anomaly detection in high-dimensional unstructured data. Engineered a multi-modal feature set by integrating signal intensity, fragment size distribution, and sequence diversity metrics. Managed the full data lifecycle from raw ingestion to model inference, delivering a robust computational tool that significantly improved prediction accuracy over existing baselines.
2. [PMID: 37942753](#) | Lead a data innovation initiative that disproved long-standing assumptions regarding data quality in legacy systems. Leveraged novel data capture technologies to uncover previously overlooked patterns in high-volume datasets. Owned the analytical strategy from concept to publication, delivering critical data-driven insights that redefined the team's approach to signal extraction.
3. [PMID: 39262778](#) | Executed a high-impact validation study that reshaped the organization's data acquisition strategy. Through A/B testing and rigorous hypothesis validation, identified that specific data subsets provided no additional predictive value. These findings prevented resource waste on low-yield data channels and redirected R&D efforts toward more profitable high-signal data sources.

Full publication list: [Scopus](#) | [Google Scholar](#) | [ORCID](#)

Grants and prizes

2024 | Dutch Research Council (NWO) | Augmenting Datasets using Diffusion Models for Liquid Biopsy Analysis.

2023 | The Netherlands Organisation for Health Research and Development (ZonMw) | Full-length Circular RNA Detection in Liquid Biopsies for Cervical Cancer Detection.

2018 | Hungarian Scientific Research Fund (OTKA) | Multiplatform analysis of herpesvirus gene expression.

2018 | Hungarian Scientific Research Fund (OTKA) | Using OMICS approach for the analysis of viral gene expression.

2018 | National Excellence Programme (UNKP) | Epstein-Barr virus and Host Cell Transcriptome Analysis

2017 | National Excellence Programme (UNKP) | AcMNPV Transcriptome Analysis with Long-Read Sequencing.

Selected packages

1. [FrEIA](#) | Open-source Python library designed for high-dimensional feature extraction. Implements modular algorithms to parse unstructured sequence data into structured metrics (dimensionality reduction, signal diversity) for downstream ML applications.
2. [ITSFASTR](#) | Scalable data processing suite for high-throughput signal analysis. Orchestrates the cleaning, quality control, and statistical analysis of massive datasets, optimized to handle the distinct error profiles of different data generation technologies.
3. [Ouro-tools](#) | Specialized toolkit for non-linear data structures. A preprocessing engine designed to handle topological complexities (circularity) in raw data, implementing advanced denoising and alignment algorithms to prepare data for production pipelines.

